



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to AMMAN ENTERPRISE

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

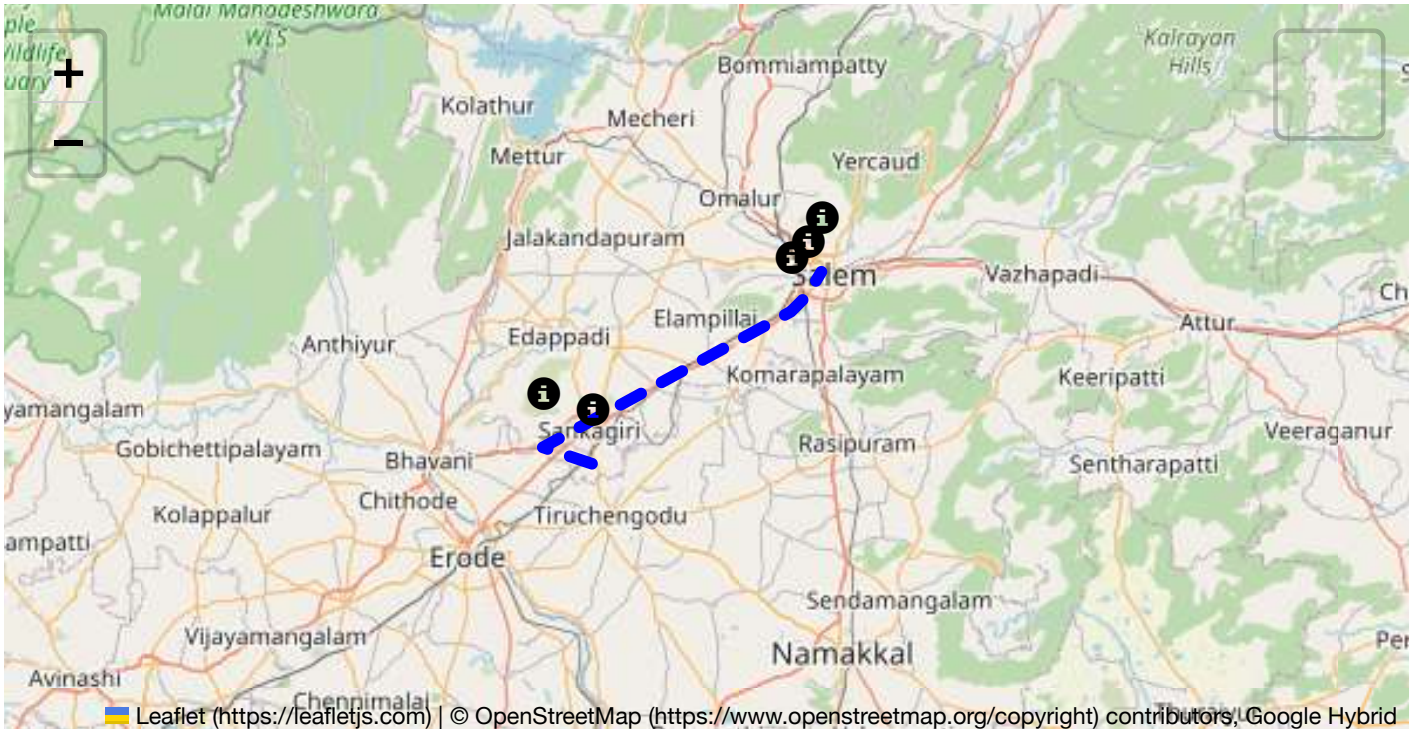
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

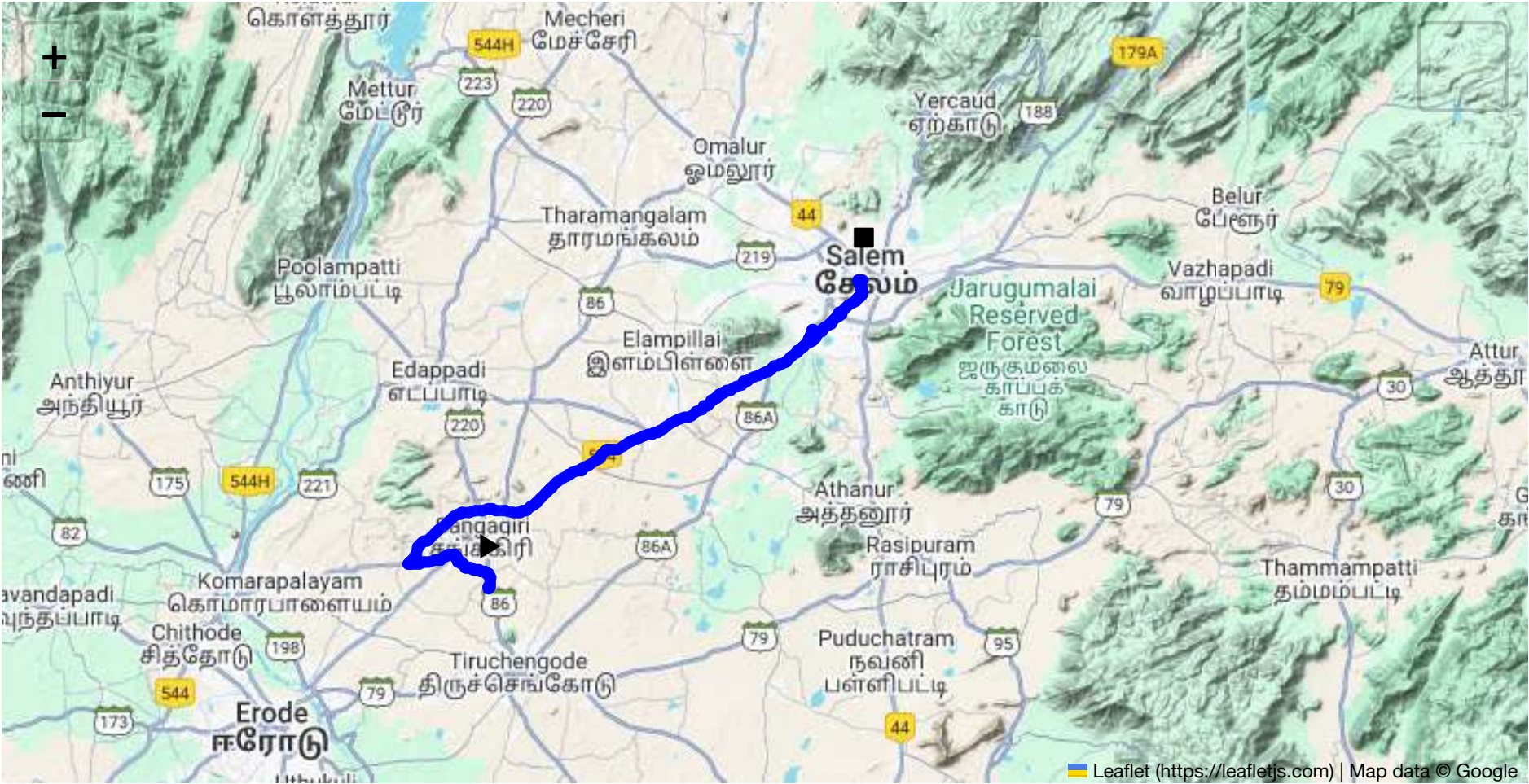
This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 56.16 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.5 hours

Start: (11.4381, 77.8734)

End: (11.656488, 78.142601)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Leigh Bazar Rd, Salem, Tamil Nadu passes through several key junctions and urban areas. It begins at Sangagiri, moving through small towns like Padaiveedu, and follows major roads such as Bhavani Main Road and Coimbatore Road as it approaches Salem.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region typically experiences a tropical climate. The summer months (March to June) can be very hot, increasing pavement temperatures, which may contribute to tire blowouts or road fatigue. Monsoon season (June to September) brings heavy rain which can cause flooding and reduced visibility. Winters (December to February) are mild, with fewer weather-related hazards.

3. Analysis of Traffic Patterns

Traffic can peak during morning (8-10 AM) and evening hours (5-8 PM), particularly on the approach to Salem. Congestion-prone areas include urban sections near major intersections and market areas in Kondalampatti and Salem, which should be considered in route planning.

4. Assessment of Road Quality and Infrastructure

The road quality varies; state highways and main roads are generally well-maintained, but secondary roads may have potholes or uneven surfaces. Pay attention to construction zones, as ongoing infrastructure projects can affect travel times and road safety.

5. Suggestions for Alternative Routes for Emergencies

An alternative route could involve taking NH44 from Sangagiri direct to Salem by bypassing smaller towns using Kolathur Road, which may offer better conditions during emergencies. However, local conditions should be monitored as factors like roadworks or accidents can influence viability.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transport of hazardous materials in Tamil Nadu requires compliance with road regulations, including proper labeling and safety documentation. Specific restrictions apply within urban areas of Salem, especially around densely populated areas.

7. Overview of Historical Incidents

There have been occasional reports of vehicle accidents involving heavy trucks in the region, often related to adverse weather or human error. Past incidents highlight the importance of speed regulation and caution during wet conditions.

8. Environmental Considerations and Sensitive Areas

The route passes through agricultural areas which are sensitive to environmental disturbances. Dust and chemical spills can have detrimental effects, so environmental precautions are necessary during transport.

9. Analysis of Communication Coverage

Mobile network coverage is generally reliable on main roads, but there may be occasional dead zones in rural stretches, particularly in less populated areas. Planning for communications is advisable.

10. Estimated Emergency Response Times

In rural areas, response times can be longer, potentially over 30 minutes, due to distance from emergency services. In urban areas near Salem, response times are typically 10-20 minutes thanks to closer proximity to facilities.

12. Overall Summary of Risk Assessment

This route, while mostly on primary roads, poses moderate risks related to weather, road congestion, and infrastructure variability. Drivers should be aware of peak traffic times and adhere to safety regulations strictly. Emergency communication plans and alternative routing should be part of logistical planning to mitigate potential transportation hazards.

Overall, while the route presents some challenges typical for heavy vehicle operation, proper planning and compliance with regulations can provide a safe journey for hazardous material transport. Regular updates on road conditions and local advisories are recommended to stay informed of any changes.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Roundabout	High	11.63266, 78.12699	15 KM/Hr
2	U-Turn	High	11.6302987, 78.12452429999999	10 KM/Hr
3	Turn	Medium	11.43822, 77.87348	30 KM/Hr
4	Turn	High	11.43968, 77.87345	15 KM/Hr
5	Turn	High	11.44029, 77.87544	15 KM/Hr
6	Turn	Medium	11.45358, 77.85781	30 KM/Hr
7	Turn	Medium	11.45353, 77.85697	30 KM/Hr
8	Turn	Medium	11.45839, 77.85143	30 KM/Hr
9	Turn	Medium	11.46053, 77.85034	30 KM/Hr
10	Turn	Medium	11.46307, 77.84941	30 KM/Hr
11	Turn	Medium	11.46316, 77.84921	30 KM/Hr
12	Turn	High	11.45542, 77.81725	15 KM/Hr
13	Blind Spot	Blind Spot	11.45631, 77.81668	10 KM/Hr
14	Turn	Medium	11.60942, 78.10056	30 KM/Hr
15	Turn	Medium	11.60965, 78.10070	30 KM/Hr
16	Blind Spot	Blind Spot	11.62145, 78.10740	10 KM/Hr
17	Turn	High	11.62130, 78.10776	15 KM/Hr
18	Blind Spot	Blind Spot	11.61217, 78.10439	10 KM/Hr
19	Turn	High	11.63306, 78.12722	15 KM/Hr
20	Turn	Medium	11.63294, 78.12752	30 KM/Hr
21	Turn	Medium	11.63283, 78.12757	30 KM/Hr
22	Turn	High	11.63262, 78.12754	15 KM/Hr
23	Turn	Medium	11.63026, 78.12481	30 KM/Hr
24	Turn	Medium	11.63025, 78.12470	30 KM/Hr
25	Blind Spot	Blind Spot	11.63018, 78.12462	10 KM/Hr
26	Blind Spot	Blind Spot	11.63030, 78.12452	10 KM/Hr
27	Blind Spot	Blind Spot	11.63018, 78.12462	10 KM/Hr
28	Turn	High	11.63028, 78.12474	15 KM/Hr
29	Turn	High	11.63026, 78.12481	15 KM/Hr
30	Turn	Medium	11.64854, 78.14335	30 KM/Hr
31	Turn	Medium	11.65152, 78.14229	30 KM/Hr
32	Blind Spot	Blind Spot	11.65573, 78.13972	10 KM/Hr
33	Turn	High	11.65623, 78.14218	15 KM/Hr
34	Turn	High	11.65658, 78.14217	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
14	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
16	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
17	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
18	clinic	Salem Care Foundation	11.6371946, 78.1313713	30 km/h	Medium
19	hospital	Aravind Eye Hospital	11.641927, 78.1392347	30 km/h	Medium
20	police	Annadanapatti B 4 Police Station	11.6424031, 78.1404779	30 km/h	Medium
21	clinic	Shri Bharthi Nursing Home	11.6420069, 78.1424099	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
10	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
11	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
12	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
15	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63266, 78.12699



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63266, 78.12699



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 11.6302987, 78.12452429999999



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45358, 77.85781



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45839, 77.85143



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46053, 77.85034



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46316, 77.84921



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60942, 78.10056



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60965, 78.10070



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.62145, 78.10740



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.62130, 78.10776



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61217, 78.10439



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63306, 78.12722



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63294, 78.12752



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63283, 78.12757



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63262, 78.12754



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63026, 78.12481



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63025, 78.12470



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.63018, 78.12462



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.63030, 78.12452



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.63018, 78.12462



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63028, 78.12474



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63026, 78.12481



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64854, 78.14335



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65152, 78.14229



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.65573, 78.13972



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65623, 78.14218



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65658, 78.14217
